```
#include <iostream>
#include <climits>
using namespace std;

#define V 5

void primMST(int graph[V][V])
{
   int parent[V];
   int key[V];
   bool visited[V] = {false};

   for (int i = 0; i < V; i++)
      key[i] = INT_MAX;

   key[0] = 0;
   parent[0] = -1;

   for (int count = 0; count < V - 1; count++)
   {
      int min = INT_MAX;
      int u = -1;

      for (int v = 0; v < V; v++)
      {
         if (!visited[v] && key[v] < min)
         {
            min = key[v];
```

```
            u = v;
        }
    }

    if (u == -1)
    {
        cout << "Network is disconnected!\n";
        return;
    }

    visited[u] = true;

    for (int v = 0; v < V; v++)
    {
        if (graph[u][v] != 0 &&
            !visited[v] &&
            graph[u][v] < key[v])
        {
            parent[v] = u;
            key[v] = graph[u][v];
        }
    }
}

int totalCost = 0;

cout << "\nMinimum Cost Communication Network\n";
cout << "Connection\tInstallation Cost\n";
```

```
    for (int i = 1; i < V; i++)
    {
        cout << "Block " << parent[i] + 1
             << " - Block " << i + 1
             << "\tRs. " << graph[i][parent[i]] << endl;

        totalCost += graph[i][parent[i]];
    }

    cout << "\nMinimum Total Installation Cost: Rs. "
         << totalCost << endl;
}

int main()
{
    int graph[V][V] = {
        {0, 10, 0, 30, 0},
        {10, 0, 20, 0, 50},
        {0, 20, 0, 40, 10},
        {30, 0, 40, 0, 20},
        {0, 50, 10, 20, 0}
    };

    cout << "Hospital Communication Network\n";
    cout << "Blocks: ICU, Lab, Emergency, Admin, Pharmacy\n";

    primMST(graph);
```

```
    return 0;
}
```